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Mobile Survey Design Research
Experiment 27: Breadcrumb Trail

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Abstract

User interface (UI) breadcrumbs are a navigational component that enables users to be aware of their current location within the hierarchical structure of information source (e.g., a website) and to trace back to a higher level of the information architecture. However, it is yet to know if respondents can benefit from having breadcrumbs available on the screen in terms of reducing respondents' burden and improving data quality. The present study compared usability between two navigation designs: With breadcrumb trail and without breadcrumb trail. The findings show that few participants in the breadcrumb group used the breadcrumb trail and there was no difference in task completion rate between the two designs. Participants navigated through survey pages faster with no-breadcrumb trail in one type of task but not the other type. Participants rated surveys using either design as easy to complete.

Keywords: Mobile Survey, Breadcrumb Trail, Navigation Design

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Introduction

1. Purpose

User interface (UI) breadcrumbs are a navigational component that enables users to be aware of their current location within the hierarchical structure of information source (e.g., a website) and to trace back to a higher level of the information architecture. In some surveys with complex navigation design, using breadcrumbs to facilitate navigation would be a design option. Due to a smaller display area on a smartphone, it is important to understand how much respondents can benefit from having breadcrumbs available on the screen in terms of reducing respondents' burden and improving data quality. By characterizing respondents' behavior with regards to breadcrumbs, we can be informed about the usefulness of breadcrumbs in mobile survey instrument. The purpose of this study was to investigate if breadcrumb trails improve navigation performance on an online survey using a smartphone. A recommendation on the use of breadcrumbs will be based on whether navigation performance is improved with using breadcrumbs.

2. Specific Research Question

When completing a survey on a smartphone, do respondents carry out a navigation task more accurately, faster, with better satisfaction, with the presence of a breadcrumb trail on the survey screen than without?

Methods

1. Participants

Thirty-seven participants (Ps) (31 females, 6 males) participated in the study. Detailed demographics are shown in Table 1. Seventy-eight percent of participants reported using a smartphone for more than two years, with 14 percent reporting using a smartphone less than 2 years, but more than one year and 8 percent reporting using a smartphone between six months and one year. On a scale of 1-5, where higher scores represent more smartphone familiarity, participants reported an average of 3.84 ($SD = 1.01$). The participants had habitual near vision of 20/50 or better (sufficient for reading newspaper) and had no color vision impairment.

All participants received a Privacy Act Statement that states the purpose of the study, information to be collected, and the authority under which the information is collected. A written consent to data collection was obtained from all participants. The data collection was approved by the U.S. Office of Management and Budget. All participants were from the metropolitan area of Washington DC and were recruited at community centers.

Table 1. Participants Demographics

Gender (number of participants)		
Male		6
Female		31
Age		
Mean (SD)		66.9 (5.4)
Range		60 - 81
Education (number of participants)		

Less than high school diploma		4
Completed high school		4
Some college, no degree		8
Associate degree		7
Bachelor degree		5
Post graduate degree		9
Hispanic origin (number of participants)		
Yes		9
No		27
No Response (N/A)		1
Race (number of participants)		
White		18
Black or African American		10
Asian		5
Mixed Races		1
Missing		3

2. Experimental Paradigm

Each participant completed a survey on an iPhone using dummy information. The flow of the survey is shown in Figure 1. The participant was instructed to notify the test administrator (TA) when they saw a Stop notice on the screen (the 5th screen on the top row and the 3rd on the bottom row in Figure 1) during survey completion. At each Stop, the participant performed a task. *Task 1*: At the first stop, the participant was asked to return to the Household section (the first screen on the top row in Figure 1). *Task 2*: At the second stop, the participant was asked to return to John Doe's section (the 6th screen on the middle row in Figure 1). This paradigm was designed to simulate the scenario of a household census survey in which the same set of questions were iterated to enumerate all members in the same household. A breadcrumbs trail may be useful for finding a way back to the household roster.

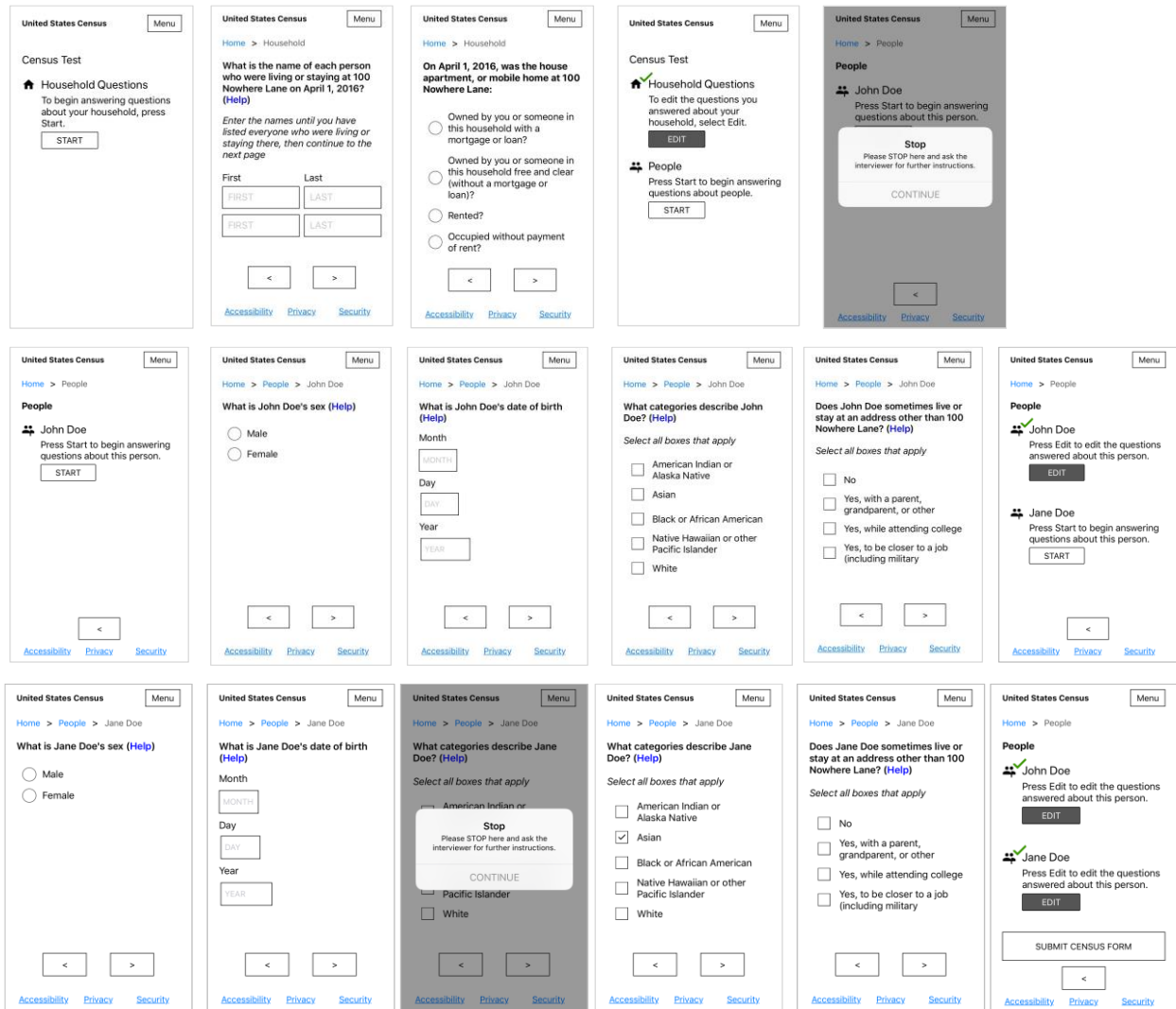


Figure 1. The flow of the survey, starting from the top-left screen to the bottom-right screen.

3. Screen Design

Two types of screens were designed. One design had a location-based breadcrumb trail (Figure 2, left) while the other had no breadcrumb trail (Figure 2, right).

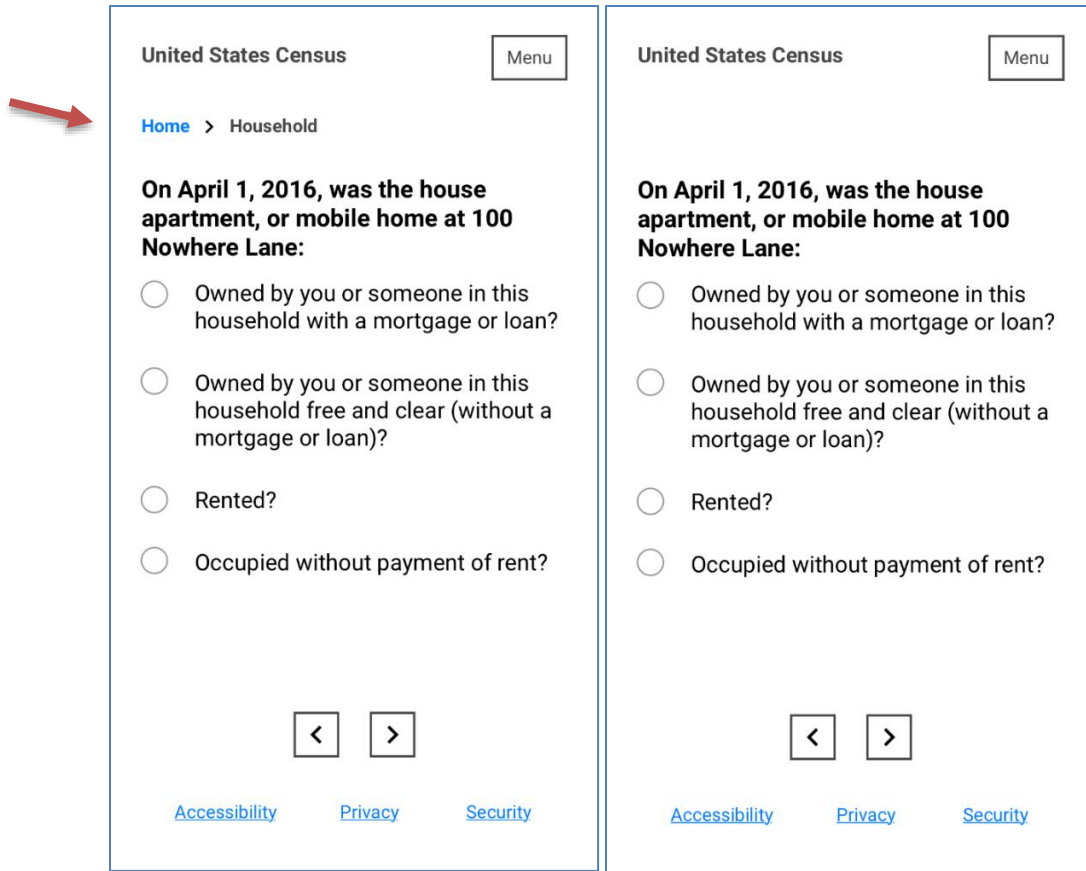


Figure 2. A screen design of breadcrumb trail (left, arrow pointing at trail), and no breadcrumb (right).

4. Performance Measures

a. *Navigational Efficiency:*

- i. *Task Completion Time:* Length of time that it takes the participant to navigate from the task start page (Stop screen) to the page of the correct section to make the instructed change (e.g., in Figure 1, the time between the fifth screen in the top row and (going backwards) the first screen in the top row).
- ii. *Breadcrumb link used:* Yes or no.

b. *Effectiveness:*

- i. *Task success:* Success - The participant reaches the edit button for the correct section. Fail - P, in Task 1, navigates forward (instead of backwards) and reaches the beginning of the next task; or in Task 2, navigates forward (instead of backwards) and completes the survey.

c. *Satisfaction:*

- i. *Difficulty rating:* Self-report of task difficulty on a 5-point scale with 1 being very easy and 5 being very difficult.

5. Experimental Factor

Breadcrumb Presence (2 levels)

Level 1: No breadcrumb trail (no-breadcrumb)

Level 2: Breadcrumb trail (breadcrumb)

6. Experimental Design

This was a between-subjects design with a single experimental factor of 2 levels, no breadcrumb trail vs. breadcrumb trail. Participants were randomly assigned to either condition, with 19 in the breadcrumb group and 18 in the no-breadcrumb group.

7. Experimental Procedure

- a. Participants completed the survey using dummy information and performed the two tasks described in the Experimental Paradigm section.
- b. After the survey was complete, participants rated the ease of completing the task on a 5-point Likert scale where '1' is *very easy* and '5' is *very difficult*.
- c. Participants were shown an example of both conditions, breadcrumb present and breadcrumb absent, on a laptop and asked to choose which version they prefer.

8. Data Analysis

Since only two of 19 participants in the Breadcrumb group actually used the breadcrumb trail, navigation performance of using a breadcrumb trail could not be meaningfully assessed. We thus decided to investigate the effect of breadcrumb presence on the mobile survey screen on navigation performance in participants who did not use the breadcrumb feature ($N = 17$). We conducted a Welch's t -test to determine the difference in task completion time between the two conditions, a binary logistic regression predicting task accuracy from condition, and a multinomial logistic regression predicting satisfaction from condition.

Results

Use of breadcrumb: Two out of 19 participants in the breadcrumb group used breadcrumb trail. The following results compared the participants who did not use breadcrumb trail in the breadcrumb group and those in the no-breadcrumb group.

Effectiveness: In the breadcrumb group, 68.4% of Ps completed Task 1 successfully and 70.6% Task 2. In the no-breadcrumb group, 72.2% of Ps completed Task 1 successfully and 77.8% Task 2. The results of a binomial logistic regression predicting task success from condition (breadcrumb vs. no-breadcrumb) show no statistically significant effect of condition on accuracy for either task ($p > .05$).

Efficiency: Due to unequal variances between the two groups, a *Welch's t*-test was conducted to assess the effect of breadcrumb presence on task completion time. There was a marginally significant difference in task completion time for Task 1 (*Welch's t*(1, 24.25) = 3.87, $p = .06$), with shorter completion time in the no-breadcrumb group ($M = 40.0 \pm 44.8$ sec) compared to breadcrumb group ($M = 93.1 \pm 108.3$ sec). Looking at Task 2 alone, there

was no statistically significant difference in completion time between the two groups ($M = 25.0 \pm 22.0$ sec (no-breadcrumb) vs. 32.4 ± 40.7 sec (breadcrumb); $p > .05$).

Satisfaction: Participants rated the task as easy overall, with over 70% giving either a 1 or 2 (lowest difficulty levels). See Table 2. Results of a multinomial logistic regression predicting difficulty rating from condition show no statistically significant effect of condition on difficulty ratings ($p > .05$).

Table 2. Distribution of difficulty ratings by condition (Count, Percentage)

Condition	Difficulty Scale				
	1	2	3	4	5
<i>Breadcrumb</i>	8, 47.1%	5, 29.4%	2, 11.8%	2, 11.8%	0, 0%
<i>No breadcrumb</i>	11, 61.1%	4, 22.2%	3, 16.7%	0, 0%	0, 0%

Recommendation

The present study compared usability between two navigation designs: with breadcrumb trail and without breadcrumb trail. The findings show that few participants in the breadcrumb group used breadcrumb trail. There was no difference in task completion rate between the two designs. Participants navigated through survey pages faster with no-breadcrumb trail in one type of task but not the other type. Participants rated surveys using either design as easy to complete. Eliminating the breadcrumb trail makes more screen estate to be used for presenting other survey information.

We recommend that no breadcrumb trails be used in mobile surveys, unless there is a compelling reason to believe that respondents will need to perform more complex navigation than is typically expected.